

Unit A2: Networks (Core)

Sub-topics: A2.1 to A2.4 (SL Content)

1. Syllabus Scope & Learning Outcomes

The "Do" column defines the practical skills required for the exam (Paper 1).

Syllabus Point	Command Term	Student Expectation (The "Do")
A2.1.1	Describe	Describe the purpose/characteristics of LAN, WAN, PAN (Bluetooth), and VPN.
A2.1.2	Describe	Describe digital infrastructures: Internet, Cloud, Edge Computing, and Mobile Networks (5G).
A2.1.3	Describe	Describe network devices: Gateway, Hardware Firewall, Modem, NIC, Router, Switch, WAP.
A2.1.4	Describe	Describe protocols: TCP (Reliable), UDP (Fast), HTTP, HTTPS, DHCP.
A2.2.1	Describe	Describe network topologies: Star (central switch), Mesh (interconnected), Hybrid.
A2.2.3	Compare	Compare Client-Server vs. Peer-to-Peer (P2P) models regarding centralization, cost, and reliability.
A2.2.4	Explain	Explain Network Segmentation using

		Subnetting and VLANs (Virtual LANs) for security/performance.
A2.3.1	Describe	Describe IP Addressing: IPv4 (32-bit) vs. IPv6 (128-bit), Public vs. Private, Static vs. Dynamic, NAT.
A2.3.2	Compare	Compare transmission media: Fibre Optic, Twisted Pair (Copper), Wireless.
A2.3.3	Explain	Explain Packet Switching: Segmentation, Packet Header (IPs, Sequence), Routing, Reassembly.
A2.4.1	Discuss	Discuss Firewall effectiveness: Inspection of packets against Allow/Deny lists (IP/Port).
A2.4.4	Describe	Describe Encryption: Symmetric (Shared Key) vs. Asymmetric (Public/Private Keys) and Digital Certificates.

2. Key Terminology & Definitions

- **LAN (Local Area Network):** Connects devices in a small geographical area (home/office). High bandwidth, low latency.
- **WAN (Wide Area Network):** Connects multiple LANs over large distances (cities/countries).
- **VPN (Virtual Private Network):** Creates a secure tunnel over a public network (internet) to a private network.
- **Edge Computing:** Processing data closer to the source (the "edge") to reduce latency (e.g., Netflix local caches).
- **Router:** Directs data packets *between* different networks based on IP addresses.
- **Switch:** Connects devices *within* a single network segment using MAC addresses.
- **Protocol:** A set of rules defining how data is transmitted and received.

- **NAT (Network Address Translation):** Translates private IP addresses to a single public IP address (conserves IPv4).
- **Packet Switching:** Breaking data into chunks that travel independently across the network.
- **Symmetric Encryption:** Uses the *same* key to encrypt and decrypt (faster, but key exchange is risky).
- **Asymmetric Encryption:** Uses a *Public Key* to encrypt and a *Private Key* to decrypt (slower, but secure key exchange).

3. Core Content & Exam Actions

A2.1: Fundamentals

Theory:

- **Protocols:**
 - **TCP:** Connection-oriented. Guarantees delivery (re-sends lost packets). Used for emails, files, web.
 - **UDP:** Connectionless. No guarantee. Faster/lower latency. Used for streaming, gaming.
- **Devices:**
 - **Modem:** Modulates/Demodulates signals (Digital <-> Analogue).
 - **Gateway:** Translates between different network protocols.

Student Exam Action:

- **Differentiation:** Clearly distinguish between a **Switch** (central hub of a Star topology) and a **Router** (gateway to the outside world).

A2.2: Architecture

Theory:

- **Topologies:**
 - **Star:** All nodes connect to a central switch. If switch fails, network fails. No collisions.
 - **Mesh:** Nodes connect directly to many others. High redundancy/reliability. Expensive cabling.
- **Segmentation:**
 - **VLAN:** Logically separating devices on the same physical hardware (e.g., separating Student and Staff traffic for security).

Student Exam Action:

- **Diagramming:** Sketch a Star topology (central box with spokes) vs. a Mesh topology (web of connections).

A2.3: Data Transmission

Theory:

- **IP Addressing:**
 - **IPv4:** 4 bytes (e.g., 192.168.1.1). Running out of addresses.
 - **IPv6:** 16 bytes (Hexadecimal). Massive address space.
- **Media Comparison:**
 - **Fibre:** Fastest, secure (hard to tap), expensive, fragile.
 - **Twisted Pair:** Cheap, easy to install, susceptible to interference (EMI).
 - **Wireless:** Flexible (mobility), slower, susceptible to interference/walls.

Student Exam Action:

- **Explanation:** Explain why Packet Switching is more efficient than Circuit Switching (lines aren't tied up; packets take optimal routes).

A2.4: Security (Core)

Theory:

- **Firewalls:** Filter traffic based on rules (Source IP, Destination Port). Can be Hardware or Software.
- **Encryption:**
 - **Diffie-Hellman:** A method to securely exchange cryptographic keys over a public channel.
 - **Digital Certificates:** Issued by a Certificate Authority (CA) to verify the owner of a Public Key (used in HTTPS).

Student Exam Action:

- **Scenario:** Given a scenario (e.g., "Alice sends a file to Bob"), identify whether Symmetric or Asymmetric encryption should be used (Symmetric for the large file speed; Asymmetric to share the key).

4. Common Pitfalls & Examiner Tips

- **Pitfall (Router vs Switch):** Students often use these interchangeably. **Switch** = Local traffic (MAC address). **Router** = Internet traffic (IP address).
- **Pitfall (Encryption):** Thinking "HTTPS" means the website is safe. It only means the *connection* is encrypted, not that the site owner is trustworthy.
- **Tip (NAT):** Always mention **two** benefits of NAT: 1. Security (hides internal IPs). 2. Conservation (uses only one Public IP).
- **Tip (VPN):** VPNs do not "make you anonymous"; they shift your trust from your ISP to the VPN provider.